

Chapter 8 — Introduction to documentation and version control

Chapters 5 through 7 made datasets cleaner, better understood, and fairer to evaluate

Learning objectives

- Connect change tracking to regulated settings such as clinical research

Why documentation enables reproducibility

- Reproducibility requires more than a table of numbers
- Others need the context of collection, processing, and intended use
- Documentation captures metadata, structure, content
- Without that context

Documentation as collaboration

- In multi-person projects, documentation is a shared communication layer
- Clear notes let collaborators grasp structure, content
- Handoffs to new members or external partners become safer when each release explains how

Example 8.1 — Undocumented Image Classification Dataset

- Example 8.1 — hands-on module
- Example 8.1 considers an animal image classification corpus without documentation
- A new user may not know whether labels distinguish dog from puppy
- The lesson is that missing context blocks reuse and invites inconsistent future work
- Explore the chapter example module
- View files: ``modules/chapter8/example1/``

Why datasets need version control

- Datasets evolve through cleaning, enrichment
- Avoid overwriting one another's work
- Tools that combine Git with large-file or data-aware layers help when binaries and shared

Example 8.2 — Feature Drift Across Dataset Versions

- Example 8.2 — hands-on module
- Example 8.2 shows a modeling dataset that gains or modifies features during development
- If a new feature later hurts model quality
- Without that trail, feature drift is hard to diagnose
- Explore the chapter example module
- View files: ``modules/chapter8/example2/``

Example 8.3 — Clinical Dataset Change Tracking for Compliance

- Example 8.3 — hands-on module
- Example 8.3 highlights regulated clinical research
- Accountability requires linking each modification to an author and a clear explanation
- Version control therefore supports both scientific reproducibility and compliance evidence
- Explore the chapter example module
- View files: ``modules/chapter8/example3/``

Takeaways

- Documentation turns raw observations into interpretable assets
- Collaboration depends on shared context and merge-safe change tracking
- Regulated domains add auditability requirements on top of everyday engineering needs

Next

- Complete the quiz for this part
- The next part moves from why documentation matters to concrete components